

RESEARCH STATEMENT

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1. INTRODUCTION

My research interests lie in differential geometry, specifically in the study of special geometric structures. These are geometric structures on manifolds that enjoy special properties, often due to them satisfying a natural partial differential equation. Because of this, their existence and properties are closely connected to many difficult questions in other fields of geometry such as gauge theory, moduli spaces, and calibrated geometry. In turn, finding examples of these structures or proving theorems that characterize when they exist on a given manifold is usually very difficult. Along these lines, most of my research experience thus far has been trying to construct new examples and find necessary conditions for such structures to exist within G_2 -geometry, which is the differential geometry of the seven-dimensional cross product.

During my undergraduate program in Pure Mathematics at the University of Waterloo, I have been blessed to have this interest fostered by my supervisor, Professor Spiro Karigiannis, whose research is in G_2 -geometry. This sparked my broader interest in special geometric structures. These interests were then deepened and entrenched in my research experiences, which I now describe.

2. RESEARCH EXPERIENCES

I am currently working in my third and final research assistantship at the University of Waterloo, as part of my degree's co-op program where I work full-time for six terms of my five-year (fourteen term) program. Altogether, these experiences have wholly convinced me that I want to pursue mathematics research as a career.

My second research assistantship was with Professor Yizhou Zhang in Waterloo's School of Computer Science in the theory of programming languages with probabilistic constructions, which I initially decided to pursue to determine if I would prefer computer science research over math given my programming background. This ended up not being the case. Unfortunately, the assistantship ended halfway through due to an unforeseen medical issue. In turn, my only output of that term was a formalization (in the theorem prover Lean) of the basic machinery of quasi-Borel spaces, a categorical setting for probability theory [7], which Professor Zhang and his collaborators plan to use in future work. Aside from determining that I wanted to pursue math research, the main thing I got out of this experience was practice with automated theorem provers, which I believe will be broadly useful in the future mathematical landscape, and a much-needed solidification of my skills in measure theory and category theory.

Next, I describe my research program as it began in my first research assistantship under Professor Spiro Karigiannis at the University of Waterloo in the summer of 2024, which continued with my individual efforts for the year after, and is the focus of my current assistantship (which will last until December) under Professor Xuemiao Chen and Professor Karigiannis.

My research with Professor Karigiannis began by trying to understand similarities between symplectic geometry and G_2 -geometry, which have many surface-level similarities. First, I investigated how moment maps, a key tool from symplectic geometry, might generalize to G_2 -geometry. The two main candidates were covariant moment maps from multisymplectic geometry [3] and multi-moment maps as defined by Madsen-Swann [12, 13] building on work of

Apostolov-Salamon [1]. Unfortunately, calculations of mine showed neither approach would work for our purposes, as covariant moment maps do not in general preserve a critical linear-algebraic property in G_2 -geometry, and multi-moment maps do not behave well if the Lie groups involved in their definition are not abelian.

We were next led to explore if geometric (pre)quantization from symplectic geometry could generalize to G_2 -geometry. In geometric prequantization, one builds an object called a prequantum line bundle over a symplectic manifold, whose geometry encodes a linearization of the base symplectic geometry in a certain sense. More precisely, under mild topological assumptions on the base symplectic manifold (M, ω) , one can construct a unitary connection on the prequantum line bundle L that induces a faithful representation of Lie algebras $\mathcal{C}^\infty(M) \rightarrow \mathfrak{u}(\Gamma(L))$ that lifts the map $\mathcal{C}^\infty(M) \rightarrow \mathfrak{X}_{\text{Ham}}(M)$ assigning a function f to its Hamiltonian vector field X_f . Aside from their original motivations in physics, the main benefit of such a prequantum line bundle is that this linearization allows one to algebraically compute many things about the underlying symplectic manifold.

It was in trying to find the right analogue of prequantum line bundles for G_2 -geometry that I first truly felt the frustration of research, where calculations and heuristics of mine rejected half-a-dozen candidates that Professor Karigiannis and I came up with. However, I found I was able to fight through the feelings of getting stuck and still appreciate and enjoy the math I was doing, and ultimately always found myself continually excited for the next idea to try.

The next idea did eventually come. Instead of directly trying to come up with analogues of symplectic geometry for G_2 -geometry in finite dimensions, we were inspired by the work of Lee-Leung [11] and Brylinski [5] to try to pass from a G_2 -manifold to an associated infinite-dimensional symplectic manifold, and see what we could learn from that space.

This idea proved very exciting. Starting with a G_2 -manifold M , a natural quotient of the space of maps $S^1 \rightarrow M$ forms an infinite dimensional Kähler manifold [17], from which we learned that a prequantum line bundle over M ought to be a generalization of a line bundle called a gerbe over M . Such prequantum gerbes linearize parts of the underlying G_2 -geometry in an analogous way to the symplectic case [10].

These ideas were not novel individually (though exciting to figure out for myself), having been somewhat developed for other geometries in the multisymplectic geometry literature [2, 10, 15, 16], but not heavily considered from a G_2 -geometric perspective, which has led me to many interesting unanswered questions that I am currently investigating. Based on work of Brylinski [5] and Hitchin [8] for related geometries, I was led to try to construct gerbes that encoded new topological invariants of G_2 -manifolds, particularly ones related to special codimension-3 submanifolds in G_2 -manifolds called coassociative submanifolds. This led me to independently rediscover many of the main ideas of Oliveira's work [14], where one constructs gerbes using harmonic theory and then relates the topology of these gerbes to coassociative submanifolds of a G_2 -manifold via Hodge theory. These ideas are certainly not fully explored in this context.

Moreover, by modifying Lee-Leung's original approach, I have recently been able to find infinite-dimensional versions of G_2 -structures, a new result. Building on the work of Hitchin [9], Brylinski [5], Henrich [6], and Verbitsky [17], I am currently relating their differential geometry to the extrinsic geometry of curves in the original finite-dimensional manifold, which classical results in geometric quantization suggest should correspond to gauge-theoretic constructions encoded by gerbes similar to those in Oliveira's work. My primary medium-term goal is to understand this conjectural correspondence and recover whatever results I can, both in general and in explicit examples.

There are several reasons why I find this approach interesting.

For one, the gerbes involved in the story encode topological invariants, which according to geometric quantization, are analogous to the traditional Chern class of a line bundle (a topological invariant classifying the bundle up to isomorphism). Topological invariants of G_2 -manifolds

are few and far between, and so any new ones would in and of themselves be interesting. One interesting potential application along these lines is suggested by Oliveira [14], which I intend to pursue: can one use these methods to topologically constrain the mean curvature flow of codimension-3 submanifolds of a G_2 -manifold?

Moreover, the Chern class has many important applications in complex geometry with no known analogue in G_2 -geometry, so a direct analogue could be very important if any of these applications carry over to the G_2 -setting.

In a different direction, because these prequantum gerbes linearize certain parts of the underlying G_2 -geometry, it should be possible to use them to algebraically compute genuinely global information about G_2 -manifolds, something which is very hard to do. Unfortunately, for technical reasons related to the decomposition of forms on a G_2 -manifold it seems that this direction is only viable on noncompact manifolds.

On the infinite-dimensional side, I hope (and naively expect) we can quotient the infinite-dimensional G_2 -manifold to get a finite-dimensional G_2 -manifold, which would be a new way of constructing examples of G_2 -manifolds. If this fails, at the very least I hope to be able to quotient certain algebraic objects associated to the infinite-dimensional manifold (its Lie 2-algebra of Hamiltonian 1-forms) to recover finite-dimensional information about it, using the general machinery available for algebraic reduction within multisymplectic geometry [4].

As the above descriptions hopefully convince you of, all of these applications ought to be closely connected in ways that are not well understood; figuring out all of these connections is the current long-term goal of my research.

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